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Mounting structure for an integrated sensor-lens assembly in an image capture device

Abstract

A method of assembling an image capture device that includes: positioning a first sealing member between a front housing portion and a mounting structure; connecting the mounting structure to the front housing portion such that the first sealing member forms a watertight seal therebetween; positioning a second sealing member between the mounting structure and an integrated sensor-lens assembly (ISLA); orienting the second sealing member such that a locating feature extending rearwardly from the mounting structure is aligned with a notch defined by the second sealing member to facilitate proper relative orientation of the mounting structure and the second sealing member; connecting the ISLA to the mounting structure such that the locating feature is positioned within the notch and the second sealing member forms a watertight seal between the ISLA and the mounting structure; and connecting a rear housing portion to the front housing portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 18/199,720, filed May 19, 2023, which is a continuation of U.S.

patent application Ser. No. 17/866,626, filed Jul. 18, 2022, now U.S. Pat. No. 11,696,008, which is a continuation of U.S. patent application Ser. No. 16/836,216, filed Mar. 31, 2020, now U.S. Pat. No. 11,425,286, the entire disclosures of which are hereby incorporated by reference.

TECHNICAL FIELD

(1) The present disclosure relates generally to image capture devices and, more specifically, to a housing assembly for such devices.

BACKGROUND

(2) Image capture devices are used in various applications, including, for example, handheld cameras and video recorders, drones, etc. Image capture devices typically include one or more optical elements (e.g., lenses) that capture content by receiving and focusing light as well as one or more image sensors that convert the captured content into an electronic image signal that is processed by an image signal processor to form an image. In some image capture devices, the optical element(s) and the image sensors are integrated into a single unit known as an integrated sensor-lens assembly (ISLA). In addition to the ISLA, image capture devices include an array of internal components, which can result in a cumbersome, complex assembly process.

(3) To address this issue, the present disclosure describes an improved housing assembly for image capture devices that not only simplifies assembly, but provides a watertight internal environment.

SUMMARY

(4) In one aspect of the present disclosure, a housing assembly for an image capture device is disclosed. The housing assembly includes: a front housing portion defining an opening; a rear housing portion that is connected (secured) to the front housing portion so as to form a watertight seal therebetween; a mounting structure that is connected (secured) to the front housing portion adjacent to the opening; a first sealing member that is positioned between the mounting structure and the front housing portion and configured to form a watertight seal therebetween; an integrated sensor-lens assembly (ISLA) that is connected (secured) to the mounting structure such that the ISLA extends through the opening in the front housing portion; and a second sealing member that is positioned between the ISLA and the mounting structure and configured to form a watertight seal therebetween. In certain embodiments, the mounting structure may be configured as a discrete component that is separate from the front housing portion and the rear housing portion. In certain embodiments, the front housing portion may define a tongue and the rear housing portion may define a recess (e.g., a groove) that is configured to receive the tongue. In certain embodiments, the housing assembly may include an adhesive to secure the tongue within the recess to thereby form the watertight seal between the front housing portion and the rear housing portion. In certain embodiments, the front housing portion may include a flange defining the opening. In certain embodiments, the mounting structure may be connected (secured) to the flange. In certain embodiments, the flange may define a channel that is configured to receive the first sealing member. In certain embodiments, the first sealing member may include a first configuration and the second sealing member may include a second configuration that is different from the first configuration. In certain embodiments, the mounting structure may include first openings that are configured to receive first fasteners to secure the mounting structure to the front housing portion and second openings that are configured to receive second fasteners to secure the ISLA to the mounting structure. In certain embodiments, the second sealing member may include a rib and feet that extend radially outward from the rib. In certain embodiments, the feet may be spaced circumferentially from each other so as to define windows therebetween that are configured to receive the second fasteners such that the second fasteners are extendable through the ISLA and the second sealing member into the mounting structure. In certain embodiments, the mounting structure may be configured for releasable connection to a cover that is configured to conceal the ISLA, whereby the mounting structure facilitates alignment between the cover and the ISLA and between the cover and the front housing portion. In certain embodiments, the housing assembly

may further include a third sealing member that is positioned between the cover and the mounting structure and configured to form a watertight seal therebetween. In certain embodiments, the mounting structure may include a collar and ears that extend radially outward from the collar. In certain embodiments, the third sealing member may be received by the collar. It is envisioned that the housing assembly for an image capture device described above may include any combination of the features and the elements described in this paragraph.

(5) In another aspect of the present disclosure, a housing assembly for an image capture device is disclosed. The housing assembly includes: a front housing portion; a rear housing portion that is connected (secured) to the front housing portion so as to form a watertight seal therebetween; a mounting structure that is removably connected (secured) to the front housing portion; and an integrated sensor-lens assembly (ISLA) that is connected (secured) to the mounting structure. In certain embodiments, the ISLA may be directly connected to the mounting structure. In certain embodiments, the housing assembly may further include a heatsink. In certain embodiments, the ISLA may extend through the heatsink. In certain embodiments, the mounting structure may be formed as a discrete component that is separate from the heatsink. In certain embodiments, the mounting structure may be sealingly connected to both the front housing portion and the ISLA. In certain embodiments, the housing assembly may further include a first sealing member that is positioned between the front housing portion and the mounting structure and a second sealing member that is positioned between the mounting structure and the ISLA. In certain embodiments, the first sealing member and the second sealing member may have different configurations. It is envisioned that the housing assembly for an image capture device described above may include any combination of the features and the elements described in this paragraph.

(6) In another aspect of the present disclosure, a method of assembling an image capture device is disclosed. The method includes: connecting a mounting structure to a front housing portion of the image capture device such that a watertight seal is formed therebetween; positioning a heatsink within the front housing portion; connecting an integrated sensor-lens assembly (ISLA) to the mounting structure such that a watertight seal is formed therebetween; and connecting a rear housing portion of the image capture device to the front housing portion such that a watertight seal is formed therebetween. In certain embodiments, the method may further include positioning a first sealing member between the front housing portion and the mounting structure and positioning a second sealing member between the mounting structure and the ISLA. In certain embodiments, connecting the ISLA to the mounting structure may include inserting the ISLA through a corresponding opening in the heatsink such that the ISLA extends through the heatsink. It is envisioned that the method of assembling an image capture device described above may include any combination of the features and the elements described in this paragraph.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

(2) FIGS. 1A-1B are isometric views of an example of an image capture device.

(3) FIGS. 2A-2B are isometric views of another example of an image capture device.

(4) FIG. 3 is a block diagram of electronic components of an image capture device.

(5) FIG. 4 is a front, perspective view of a housing assembly for an image capture device including a front housing portion and a rear housing portion.

(6) FIG. 5 is a partial, front, perspective view of the housing assembly seen in FIG. 4 with parts

separated.

(7) FIG. 6 is a partial, rear, perspective view of the housing assembly seen in FIG. 4 with parts separated.

(8) FIG. 7 is a rear, perspective view of the front housing portion.

(9) FIG. 8 is a front, perspective view of the rear housing portion.

(10) FIG. 9 is a rear, perspective view of the front housing portion (upon assembly of the image capture device) with the rear housing portion removed.

(11) FIG. 10 is a front, perspective view of the rear housing portion (upon assembly of the image capture device) with the front housing portion removed.

(12) FIG. 11 is a rear, perspective view of a mounting structure that is configured for connection to the front housing portion.

(13) FIG. 12 is a rear, perspective view of the front housing portion illustrating the connection of an integrated sensor-lens assembly (ISLA) to the front housing portion via the mounting structure seen in FIG. 11.

(14) FIG. 13 is a front, perspective view of the housing assembly illustrating a removable cover configured to protect and conceal a lens of the ISLA and a corresponding sealing member.

(15) FIGS. 14 and 15 are front, perspective view of the housing assembly illustrating a plurality of internal sealing members.

(16) FIG. 16 is a front, perspective view of the housing assembly illustrating sealing of a front display.

(17) FIG. 17 is a rear, perspective view of the housing assembly illustrating sealing of a rear display.

DETAILED DESCRIPTION

(18) The present disclosure describes various embodiments of housing assemblies for image capture devices. The housing assemblies described herein simplify the assembly process by reducing the number of required connections and reducing (if not entirely eliminating) blind steps while improving positional accuracy (alignment) of the components of the image capture device to refine operability and enhance the overall fit and finish.

(19) The presently disclosed housing assemblies include: a front housing portion; a rear portion that is connected (secured) to the front housing portion so as to form a watertight seal therebetween; a mounting structure (e.g., a bayonet); an integrated sensor-lens assembly (ISLA); and a plurality of sealing members. More specifically, the housing assemblies described herein include a first sealing member that is positioned between the front housing portion and the mounting structure and a second sealing member that is positioned between the mounting structure and the ISLA. The sealing members facilitate the formation of watertight seals between the front housing portion, the mounting structure, and the ISLA and eliminate any need to form a seal directly between the ISLA and the front housing portion, which simplifies the assembly process. The mounting structure is configured as a discrete component that is separate from the front housing portion, the rear housing portion, and the heatsink of the image capture device.

(20) During assembly, the mounting structure is initially connected to the front housing portion with the first sealing member being positioned therebetween. Thereafter, a series of ancillary components (e.g., the light pipe, the drain cover, etc.) and the heatsink are connected to the front housing portion so as to permit connection of the GPS board, the main board, the battery cage, etc., to the heatsink as well as the establishment of any other necessary electrical connections. The ISLA is then passed through an opening in the heatsink and connected to the mounting structure such that the second sealing member is positioned between the ISLA and mounting structure. A rear display is then connected (secured) to the rear housing portion and the heatsink before adhesively securing the rear housing portion to the front housing portion. The front and rear housing portions, which are configured for engagement in a tongue-and-groove fashion, are then connected using an adhesive to establish a watertight internal environment for the components of the image capture device and

thereby inhibit (if not entirely prevent) the entry of water, debris, etc.

(21) The method of assembly facilitated by the configurations of the front and rear housing portions, the mounting structure, and the ISLA allows for connection of the ISLA to the front housing portion via the mounting structure. The connection of the ISLA to the front housing portion established by the mounting structure improves the relative positional accuracy of the ISLA, the front and rear housing portions, and the removable cover that conceals and protects the lens of the ISLA, which enhances the fit and finish of the image capture device upon assembly (and the overall aesthetic appearance) as well as sealing of the image capture device. Moreover, the architecture of the housing assembly described herein allows for a majority of the components of the image capture device to be assembled into and connected to the front housing portion (either directly or indirectly), rather than dividing assembly and connection between the front and rear housing portions. Concentrating assembly and connection to the front housing portion improves visibility, which allows for more precise placement and connection of the various components of the image capture device and more efficient testing.

(22) FIGS. 1A-B are isometric views of an example of an image capture device **100**. The image capture device **100** may include a body **102**, a lens **104** structured on a front surface of the body **102**, various indicators on the front surface of the body **102** (such as light-emitting diodes (LEDs), displays, and the like), various input mechanisms (such as buttons, switches, and/or touch-screens), and electronics (such as imaging electronics, power electronics, etc.) internal to the body **102** for capturing images via the lens **104** and/or performing other functions. The lens **104** is configured to receive light incident upon the lens **104** and to direct received light onto an image sensor internal to the body **102**. The image capture device **100** may be configured to capture images and video and to store captured images and video for subsequent display or playback.

(23) The image capture device **100** may include an LED or another form of indicator **106** to indicate a status of the image capture device **100** and a liquid-crystal display (LCD) or other form of a display **108** to show status information such as battery life, camera mode, elapsed time, and the like. The image capture device **100** may also include a mode button **110** and a shutter button **112** that are configured to allow a user of the image capture device **100** to interact with the image capture device **100**. For example, the mode button **110** and the shutter button **112** may be used to turn the image capture device **100** on and off, scroll through modes and settings, and select modes and change settings. The image capture device **100** may include additional buttons or interfaces (not shown) to support and/or control additional functionality.

(24) The image capture device **100** may include a door **114** coupled to the body **102**, for example, using a hinge mechanism **116**. The door **114** may be secured to the body **102** using a latch mechanism **118** that releasably engages the body **102** at a position generally opposite the hinge mechanism **116**. The door **114** may also include a seal **120** and a battery interface **122**. When the door **114** is in an open position, access is provided to an input-output (I/O) interface **124** for connecting to or communicating with external devices as described below and to a battery receptacle **126** for placement and replacement of a battery (not shown). The battery receptacle **126** includes operative connections (not shown) for power transfer between the battery and the image capture device **100**. When the door **114** is in a closed position, the seal **120** engages a flange (not shown) or other interface to provide an environmental seal, and the battery interface **122** engages the battery to secure the battery in the battery receptacle **126**. The door **114** can also have a removed position (not shown) where the entire door **114** is separated from the image capture device **100**, that is, where both the hinge mechanism **116** and the latch mechanism **118** are decoupled from the body **102** to allow the door **114** to be removed from the image capture device **100**.

(25) The image capture device **100** may include a microphone **128** on a front surface and another microphone **130** on a side surface. The image capture device **100** may include other microphones on other surfaces (not shown). The microphones **128**, **130** may be configured to receive and record audio signals in conjunction with recording video or separate from recording of video. The image

capture device **100** may include a speaker **132** on a bottom surface of the image capture device **100**. The image capture device **100** may include other speakers on other surfaces (not shown). The speaker **132** may be configured to play back recorded audio or emit sounds associated with notifications.

(26) A front surface of the image capture device **100** may include a drainage channel **134**. A bottom surface of the image capture device **100** may include an interconnect mechanism **136** for connecting the image capture device **100** to a handle grip or other securing device. In the example shown in FIG. **1B**, the interconnect mechanism **136** includes folding protrusions configured to move between a nested or collapsed position as shown and an extended or open position (not shown) that facilitates coupling of the protrusions to mating protrusions of other devices such as handle grips, mounts, clips, or like devices.

(27) The image capture device **100** may include an interactive display **138** that allows for interaction with the image capture device **100** while simultaneously displaying information on a surface of the image capture device **100**.

(28) The image capture device **100** of FIGS. **1A-1B** includes an exterior that encompasses and protects internal electronics. In the present example, the exterior includes six surfaces (i.e. a front face, a left face, a right face, a back face, a top face, and a bottom face) that form a rectangular cuboid. Furthermore, both the front and rear surfaces of the image capture device **100** are rectangular. In other embodiments, the exterior may have a different shape. The image capture device **100** may be made of a rigid material such as plastic, aluminum, steel, or fiberglass. The image capture device **100** may include features other than those described here. For example, the image capture device **100** may include additional buttons or different interface features, such as interchangeable lenses, cold shoes, and hot shoes that can add functional features to the image capture device **100**.

(29) The image capture device **100** may include various types of image sensors, such as charge-coupled device (CCD) sensors, active pixel sensors (APS), complementary metal-oxide-semiconductor (CMOS) sensors, N-type metal-oxide-semiconductor (NMOS) sensors, and/or any other image sensor or combination of image sensors.

(30) Although not illustrated, in various embodiments, the image capture device **100** may include other additional electrical components (e.g., an image processor, camera system-on-chip (SoC), etc.), which may be included on one or more circuit boards within the body **102** of the image capture device **100**.

(31) The image capture device **100** may interface with or communicate with an external device, such as an external user interface device (not shown), via a wired or wireless computing communication link (e.g., the I/O interface **124**). Any number of computing communication links may be used. The computing communication link may be a direct computing communication link or an indirect computing communication link, such as a link including another device or a network, such as the internet, may be used.

(32) In some implementations, the computing communication link may be a Wi-Fi link, an infrared link, a Bluetooth (BT) link, a cellular link, a ZigBee link, a near field communications (NFC) link, such as an ISO/IEC 20643 protocol link, an Advanced Network Technology interoperability (ANT+) link, and/or any other wireless communications link or combination of links.

(33) In some implementations, the computing communication link may be an HDMI link, a USB link, a digital video interface link, a display port interface link, such as a Video Electronics Standards Association (VESA) digital display interface link, an Ethernet link, a Thunderbolt link, and/or other wired computing communication link.

(34) The image capture device **100** may transmit images, such as panoramic images, or portions thereof, to the external user interface device via the computing communication link, and the external user interface device may store, process, display, or a combination thereof the panoramic images.

(35) The external user interface device may be a computing device, such as a smartphone, a tablet computer, a phablet, a smart watch, a portable computer, personal computing device, and/or another device or combination of devices configured to receive user input, communicate information with the image capture device **100** via the computing communication link, or receive user input and communicate information with the image capture device **100** via the computing communication link.

(36) The external user interface device may display, or otherwise present, content, such as images or video, acquired by the image capture device **100**. For example, a display of the external user interface device may be a viewport into the three-dimensional space represented by the panoramic images or video captured or created by the image capture device **100**.

(37) The external user interface device may communicate information, such as metadata, to the image capture device **100**. For example, the external user interface device may send orientation information of the external user interface device with respect to a defined coordinate system to the image capture device **100**, such that the image capture device **100** may determine an orientation of the external user interface device relative to the image capture device **100**.

(38) Based on the determined orientation, the image capture device **100** may identify a portion of the panoramic images or video captured by the image capture device **100** for the image capture device **100** to send to the external user interface device for presentation as the viewport. In some implementations, based on the determined orientation, the image capture device **100** may determine the location of the external user interface device and/or the dimensions for viewing of a portion of the panoramic images or video.

(39) The external user interface device may implement or execute one or more applications to manage or control the image capture device **100**. For example, the external user interface device may include an application for controlling camera configuration, video acquisition, video display, or any other configurable or controllable aspect of the image capture device **100**.

(40) The user interface device, such as via an application, may generate and share, such as via a cloud-based or social media service, one or more images, or short video clips, such as in response to user input. In some implementations, the external user interface device, such as via an application, may remotely control the image capture device **100** such as in response to user input.

(41) The external user interface device, such as via an application, may display unprocessed or minimally processed images or video captured by the image capture device **100** contemporaneously with capturing the images or video by the image capture device **100**, such as for shot framing or live preview, and which may be performed in response to user input. In some implementations, the external user interface device, such as via an application, may mark one or more key moments contemporaneously with capturing the images or video by the image capture device **100**, such as with a tag or highlight in response to a user input or user gesture.

(42) The external user interface device, such as via an application, may display or otherwise present marks or tags associated with images or video, such as in response to user input. For example, marks may be presented in a camera roll application for location review and/or playback of video highlights.

(43) The external user interface device, such as via an application, may wirelessly control camera software, hardware, or both. For example, the external user interface device may include a web-based graphical interface accessible by a user for selecting a live or previously recorded video stream from the image capture device **100** for display on the external user interface device.

(44) The external user interface device may receive information indicating a user setting, such as an image resolution setting (e.g., 3840 pixels by 2160 pixels), a frame rate setting (e.g., 60 frames per second (fps)), a location setting, and/or a context setting, which may indicate an activity, such as mountain biking, in response to user input, and may communicate the settings, or related information, to the image capture device **100**.

(45) FIGS. 2A-2B illustrate another example of an image capture device **200**. The image capture

device **200** includes a body **202** and two camera lenses **204** and **206** disposed on opposing surfaces of the body **202**, for example, in a back-to-back configuration, Janus configuration, or offset Janus configuration. The body **202** of the image capture device **200** may be made of a rigid material such as plastic, aluminum, steel, or fiberglass.

(46) The image capture device **200** includes various indicators on the front of the surface of the body **202** (such as LEDs, displays, and the like), various input mechanisms (such as buttons, switches, and touch-screen mechanisms), and electronics (e.g., imaging electronics, power electronics, etc.) internal to the body **202** that are configured to support image capture via the two camera lenses **204** and **206** and/or perform other imaging functions.

(47) The image capture device **200** includes various indicators, for example, LEDs **208**, **210** to indicate a status of the image capture device **100**. The image capture device **200** may include a mode button **212** and a shutter button **214** configured to allow a user of the image capture device **200** to interact with the image capture device **200**, to turn the image capture device **200** on, and to otherwise configure the operating mode of the image capture device **200**. It should be appreciated, however, that, in alternate embodiments, the image capture device **200** may include additional buttons or inputs to support and/or control additional functionality.

(48) The image capture device **200** may include an interconnect mechanism **216** for connecting the image capture device **200** to a handle grip or other securing device. In the example shown in FIGS. **2A** and **2B**, the interconnect mechanism **216** includes folding protrusions configured to move between a nested or collapsed position (not shown) and an extended or open position as shown that facilitates coupling of the protrusions to mating protrusions of other devices such as handle grips, mounts, clips, or like devices.

(49) The image capture device **200** may include audio components **218**, **220**, **222** such as microphones configured to receive and record audio signals (e.g., voice or other audio commands) in conjunction with recording video. The audio component **218**, **220**, **222** can also be configured to play back audio signals or provide notifications or alerts, for example, using speakers. Placement of the audio components **218**, **220**, **222** may be on one or more of several surfaces of the image capture device **200**. In the example of FIGS. **2A** and **2B**, the image capture device **200** includes three audio components **218**, **220**, **222**, with the audio component **218** on a front surface, the audio component **220** on a side surface, and the audio component **222** on a back surface of the image capture device **200**. Other numbers and configurations for the audio components are also possible.

(50) The image capture device **200** may include an interactive display **224** that allows for interaction with the image capture device **200** while simultaneously displaying information on a surface of the image capture device **200**. The interactive display **224** may include an I/O interface, receive touch inputs, display image information during video capture, and/or provide status information to a user. The status information provided by the interactive display **224** may include battery power level, memory card capacity, time elapsed for a recorded video, etc.

(51) The image capture device **200** may include a release mechanism **225** that receives a user input to in order to change a position of a door (not shown) of the image capture device **200**. The release mechanism **225** may be used to open the door (not shown) in order to access a battery, a battery receptacle, an I/O interface, a memory card interface, etc. (not shown) that are similar to components described in respect to the image capture device **100** of FIGS. **1A** and **1B**.

(52) In some embodiments, the image capture device **200** described herein includes features other than those described. For example, instead of the I/O interface and the interactive display **224**, the image capture device **200** may include additional interfaces or different interface features. For example, the image capture device **200** may include additional buttons or different interface features, such as interchangeable lenses, cold shoes, and hot shoes that can add functional features to the image capture device **200**.

(53) FIG. **3** is a block diagram of electronic components in an image capture device **300**. The image capture device **300** may be a single-lens image capture device, a multi-lens image capture

device, or variations thereof, including an image capture device with multiple capabilities such as use of interchangeable integrated sensor lens assemblies. The description of the image capture device **300** is also applicable to the image capture devices **100**, **200** of FIGS. **1A-1B** and **2A-2B**.

(54) The image capture device **300** includes a body **302** which includes electronic components such as capture components **310**, a processing apparatus **320**, data interface components **330**, movement sensors **340**, power components **350**, and/or user interface components **360**.

(55) The capture components **310** include one or more image sensors **312** for capturing images and one or more microphones **314** for capturing audio.

(56) The image sensor(s) **312** is configured to detect light of a certain spectrum (e.g., the visible spectrum or the infrared spectrum) and convey information constituting an image as electrical signals (e.g., analog or digital signals). The image sensor(s) **312** detects light incident through a lens coupled or connected to the body **302**. The image sensor(s) **312** may be any suitable type of image sensor, such as a charge-coupled device (CCD) sensor, active pixel sensor (APS), complementary metal-oxide-semiconductor (CMOS) sensor, N-type metal-oxide-semiconductor (NMOS) sensor, and/or any other image sensor or combination of image sensors. Image signals from the image sensor(s) **312** may be passed to other electronic components of the image capture device **300** via a bus **380**, such as to the processing apparatus **320**. In some implementations, the image sensor(s) **312** includes a digital-to-analog converter. A multi-lens variation of the image capture device **300** can include multiple image sensors **312**.

(57) The microphone(s) **314** is configured to detect sound, which may be recorded in conjunction with capturing images to form a video. The microphone(s) **314** may also detect sound in order to receive audible commands to control the image capture device **300**.

(58) The processing apparatus **320** may be configured to perform image signal processing (e.g., filtering, tone mapping, stitching, and/or encoding) to generate output images based on image data from the image sensor(s) **312**. The processing apparatus **320** may include one or more processors having single or multiple processing cores. In some implementations, the processing apparatus **320** may include an application specific integrated circuit (ASIC). For example, the processing apparatus **320** may include a custom image signal processor. The processing apparatus **320** may exchange data (e.g., image data) with other components of the image capture device **300**, such as the image sensor(s) **312**, via the bus **380**.

(59) The processing apparatus **320** may include memory, such as a random-access memory (RAM) device, flash memory, or another suitable type of storage device, such as a non-transitory computer-readable memory. The memory of the processing apparatus **320** may include executable instructions and data that can be accessed by one or more processors of the processing apparatus **320**. For example, the processing apparatus **320** may include one or more dynamic random-access memory (DRAM) modules, such as double data rate synchronous dynamic random-access memory (DDR SDRAM). In some implementations, the processing apparatus **320** may include a digital signal processor (DSP). More than one processing apparatus may also be present or associated with the image capture device **300**.

(60) The data interface components **330** enable communication between the image capture device **300** and other electronic devices, such as a remote control, a smartphone, a tablet computer, a laptop computer, a desktop computer, or a storage device. For example, the data interface components **330** may be used to receive commands to operate the image capture device **300**, transfer image data to other electronic devices, and/or transfer other signals or information to and from the image capture device **300**. The data interface components **330** may be configured for wired and/or wireless communication. For example, the data interface components **330** may include an I/O interface **332** that provides wired communication for the image capture device, which may be a USB interface (e.g., USB type-C), a high-definition multimedia interface (HDMI), or a FireWire interface. The data interface components **330** may include a wireless data interface **334** that provides wireless communication for the image capture device **300**, such as a Bluetooth

interface, a ZigBee interface, and/or a Wi-Fi interface. The data interface components **330** may include a storage interface **336**, such as a memory card slot configured to receive and operatively couple to a storage device (e.g., a memory card) for data transfer with the image capture device **300** (e.g., for storing captured images and/or recorded audio and video).

(61) The movement sensors **340** may detect the position and movement of the image capture device **300**. The movement sensors **340** may include a position sensor **342**, an accelerometer **344**, or a gyroscope **346**. The position sensor **342**, such as a global positioning system (GPS) sensor, is used to determine a position of the image capture device **300**. The accelerometer **344**, such as a three-axis accelerometer, measures linear motion (e.g., linear acceleration) of the image capture device **300**. The gyroscope **346**, such as a three-axis gyroscope, measures rotational motion (e.g., rate of rotation) of the image capture device **300**. Other types of movement sensors **340** may also be present or associated with the image capture device **300**.

(62) The power components **350** may receive, store, and/or provide power for operating the image capture device **300**. The power components **350** may include a battery interface **352** and a battery **354**. The battery interface **352** operatively couples to the battery **354**, for example, with conductive contacts to transfer power from the battery **354** to the other electronic components of the image capture device **300**. The power components **350** may also include the I/O interface **332**, as indicated in dotted line, and the power components **350** may receive power from an external source, such as a wall plug or external battery, for operating the image capture device **300** and/or charging the battery **354** of the image capture device **300**.

(63) The user interface components **360** may allow the user to interact with the image capture device **300**, for example, providing outputs to the user and receiving inputs from the user. The user interface components **360** may include visual output components **362** to visually communicate information and/or present captured images to the user. The visual output components **362** may include one or more lights **364** and/or more displays **366**. The display(s) **366** may be configured as a touch screen that receives inputs from the user. The user interface components **360** may also include one or more speakers **368**. The speaker(s) **368** can function as an audio output component that audibly communicates information and/or presents recorded audio to the user. The user interface components **360** may also include one or more physical input interfaces **370** that are physically manipulated by the user to provide input to the image capture device **300**. The physical input interfaces **370** may, for example, be configured as buttons, toggles, or switches. The user interface components **360** may also be considered to include the microphone(s) **314**, as indicated in dotted line, and the microphone(s) **314** may function to receive audio inputs from the user, such as voice commands.

(64) Referring now to FIGS. **4-10**, a housing assembly **400** will be discussed, which represents an alternate embodiment of the aforescribed body **102** (FIG. **1A**). More specifically, FIG. **4** provides a front, perspective view of the housing assembly **400**; FIG. **5** provides a partial, front, perspective view of the housing assembly **400** with parts separated; FIG. **6** provides a partial, rear, perspective view of the housing assembly **400** with parts separated; FIG. **7** provides a rear, perspective view of a front housing portion **402** of the housing assembly **400**; FIG. **8** provides a front, perspective view of a rear housing portion **404** of the housing assembly **400**; FIG. **9** provides a rear, perspective view of the front housing portion **402** (upon assembly of the image capture device) with the rear housing portion **404** removed; and FIG. **10** provides a front, perspective view of the rear housing portion **404** (upon assembly of the image capture device) with the front housing portion **402** removed. Although generally discussed in connection with the image capture device **100** hereinbelow, it should be appreciated that the housing assembly **400** may be configured for use with any image capture device. For example, it is envisioned that the housing assembly **400** may be adapted for use with the aforescribed image capture device **200** (FIG. **2A**) as an alternate embodiment of the body **202**.

(65) The housing assembly **400** includes the aforementioned respective front and rear housing

portions **402**, **404**; a mounting structure **500** (e.g., a bayonet **502**); a (first) sealing member **600** that is located between the mounting structure **500** and the front housing portion **402**; an integrated sensor-lens assembly (ISLA) **700**; a (second) sealing member **800** that is located between the ISLA **700** and the front housing portion **402**; a heatsink **900** that is supported by (e.g., connected to) the front housing portion **402**; and a cover **1000** (also referred to herein as a removable cover lens) that is removably connectable to (engageable with) the mounting structure **500**.

(66) The front housing portion **402** provides a framework (chassis) for various components of the image capture device **100** including, for example, the light pipe, the drain cover, various microphones, the mounting structure **500**, the ISLA **700**, the cover **1000**, and the heatsink **900**, which itself provides a framework (chassis) for various components, electronics, and circuitry that support operability and functionality of the image capture device **100**. For example, the heatsink **900** may support the display **108**, the GPS board, the main board, the battery cage, etc. The framework provided by the front housing portion **402** simplifies assembly of the image capture device **100** by focusing the area where connections are to be made and reducing (if not entirely eliminating) blind steps, which facilitates more precise alignment of the components of the image capture device **100**, as discussed in further detail below.

(67) The front housing portion **402** includes a series of internal guides **406** (FIGS. 6, 7) that facilitate assembly of the housing assembly **400**. For example, the front housing portion **402** may include a locating feature **408i** for the microphone gasket and one or more locating pins **408ii** that are configured for contact (engagement) with the heatsink **900** during connection to the front housing portion **402** to align and guide the heatsink **900** and facilitate proper location thereof within the front housing portion **402**. More specifically, the internal guides **406** are oriented along a depth *d* of the image capture device **100** such that the heatsink **900** is insertable in a forward direction. The front housing portion **402** further defines a series of mounting locations **410** (FIG. 9) that are configured in correspondence with structural supports **902** (e.g., braces **904**) included on the heatsink **900** to further facilitate secured connection of the heatsink **900** to the front housing portion **402**. For example, it is envisioned that the heatsink **900** may be connected (secured) to the front housing portion **402** via a series of mechanical fasteners (e.g., screws, pins, or the like) that are extendable through the structural supports **902** into corresponding openings defined by the front housing portion **402**.

(68) The front housing portion **402** includes a (first) opening **412** (FIG. 5), which allows for viewing of the display **108** (FIGS. 1A, 4) through the front housing portion **402**, and a receptacle **414** defining a (second) opening **416**, which receives the mounting structure **500**, the ISLA **700**, and the sealing member **800**. The opening **412** is defined by a (first) flange **418** and the opening **416** is defined by a (second) flange **420**. To protect the display **108**, it is envisioned that the image capture device **100** may include a shroud or other such covering (not shown), which may be formed as a separate, discrete component that is secured to the front housing portion **402** (e.g., to the flange **418**) in any suitable manner, such as, for example, via bonding through the use of a heat-activated adhesive. Forming the shroud (cover) as a separate, discrete component allow for the use of a variety of different materials, which reduces (if not entirely eliminates) molding defects, thereby improving the overall fit and finish of the image capture device **100**.

(69) Extending around its perimeter, the front housing portion **402** defines a planar extension (shelf) **422** and a tongue **424** that extends rearwardly (i.e., toward the rear housing portion **404**). The planar extension **422** defines a seat for the rear housing portion **404** to facilitate proper location (alignment) of the rear housing portion **404** relative to the front housing portion **402**, which further enhances the overall fit and finish of the image capture device **100** upon assembly.

(70) The rear housing portion **404** supports the interactive display **138** (FIG. 1B), as well as the circuitry and electronics supporting the operation and functionality thereof. For example, the rear housing portion **404** may include one or more flexible printed circuits (FPCs) **426** (FIG. 10), electrical connectors, the light pipe, etc., supporting imaging and tactile (touch) control of the

interactive display **138**. Additionally, it is envisioned that the rear housing portion **404** may include one or more conductive elements **428** (e.g., conductive foams) to ground the rear housing portion **404** relative to the battery of the image capture device **100**.

(71) The rear housing portion **404** includes a heat-conductive plate **430** (FIG. **8**) that allows the rear housing portion **404** to act as an additional, independent heatsink for the image capture device **100** that is separate and discrete from the heatsink **900** (FIGS. **5**, **6**). Heatsink functionality is, thus, incorporated into the rear housing portion **404**, thereby obviating the need for a separate heatsink that may otherwise be connected to the rear housing portion **404** and simplifying not only the overall construction of the image capture device **100**, but assembly as well. To facilitate the conduction of heat, the rear housing portion **404** may include (e.g., may be formed partially or entirely from) any suitable material (e.g., aluminum) or combination of materials and may be thermally connected to one or more components of the image capture device **100** (e.g., to the image sensor(s) that receive light from the lens **104** (FIG. **1A**) and/or to the image processor). It is envisioned that such thermal connections may be established in any suitable manner, such as, for example, by one or more graphite conductors. By eliminating any structural connection between the rear housing portion **404** and the heatsink **900**, the number of fasteners and connections required during assembly of the image capture device **100** can be reduced, thereby simplifying assembly.

(72) Extending around its perimeter, the rear housing portion **404** defines a recess (groove) **432** (FIGS. **8**, **10**) that is configured in correspondence with the tongue **424** (FIGS. **6**, **7**, **9**) extending from the front housing portion **402**. The recess **432** is configured to receive the tongue **424** such that the respective front and rear housing portions **402**, **404** are connectable in a tongue-and-groove fashion. To enhance the connection between the respective front and rear housing portions **402**, **404**, and further facilitate the formation of a watertight seal therebetween, it is envisioned that an adhesive **434** (FIG. **10**) may be applied to the tongue **424** and/or the recess **432** prior to insertion of the tongue **424** into the recess **432**, as described in further detail below.

(73) With reference now to FIG. **11** as well, the mounting structure **500** will be discussed. FIG. **11** provides a rear, perspective view of the mounting structure **500**, which interfaces with and/or supports support the sealing member **600**, the ISLA **700**, sealing member **800**, and the cover **1000**. The mounting structure **500** is formed as a separate, individual structure that is discrete from the front housing portion **402**, the rear housing portion **404**, the ISLA **700**, the heatsink **900**, and the cover **1000** and may include (e.g., may be formed from) any suitable material(s) of construction. To reduce (or entirely eliminate) wear on the mounting structure **500**, it is envisioned that the mounting structure **500** may include (e.g., may be formed partially or entirely from) a first material of construction and that the heatsink **900** may include (e.g., may be formed partially or entirely from) a second, different material of construction. For example, in the embodiment illustrated throughout the figures, whereas the heatsink **900** may include (e.g., may be formed partially or entirely from) aluminum, it is envisioned that that mounting structure **500** may include (e.g., may be formed partially or entirely from) steel.

(74) The mounting structure **500** is connected (secured) to the front housing portion **402** adjacent to the opening **416** (FIG. **5**) defined by the flange **420** and supports the cover **1000** and the ISLA **700**, each of which is configured for direct connection to the mounting structure **500**, as discussed in further detail below. By directly connecting the ISLA **700** to the mounting structure **500**, physical, supportive connections between the ISLA **700** and the heatsink **900** can be reduced (if not entirely eliminated), thereby simplifying assembly of the image capture device **100**. The configuration of the mounting structure **500** also eliminates any direct physical connection between the cover **1000** and the heatsink **900**, thereby simplifying sealing of the image capture device **100**. More specifically, as described in further detail below, by sealing the mounting structure **500** to the front housing portion **402** (via the sealing member **600**) and the ISLA **700** (via the sealing member **800**), sealing can be localized to the mounting structure **500**, which allows the overall geometry and

architecture of the image capture device **100** to be simplified by eliminating the need for a perimeter seal about the heatsink **900**. Moreover, the seals established between the mounting structure **500**, the front housing portion **402**, and the ISLA **700** by the sealing members **600**, **800** allows the image capture device **100** to remain watertight upon removal of the cover **1000**.

(75) The mounting structure **500** includes a (generally planar) base **504** and respective front and rear (first and second) collars **506**, **508** that extend from the base **504** in opposing (forward and rear) directions. The base **504** is configured in correspondence with the flange **420** of the front housing portion **402** such that the base **504** seats within the receptacle **414**. The base **504** includes a (central) opening **510** that is configured to receive the ISLA **700** and a series of apertures **512** that extend therethrough. The apertures **512** are positioned in alignment (registration) with corresponding apertures **436** (FIG. 5) formed in the flange **420** such that the mounting structure **500** is directly connectable to the front housing portion **402** via a series of (first) mechanical fasteners **438** (FIGS. 5, 6) (e.g., screws, pins, rivets, etc.) in either a fixed or removable fashion. Although shown as including four mechanical fasteners **438** in the illustrated embodiment, it should be appreciated that the particular number of mechanical fasteners **438** may be varied without departing from the scope of the present disclosure.

(76) To enhance the first and finish of the image capture device **100** and/or improve sealing between the mounting structure **500** and the front housing portion **402**, in certain embodiments, such as that illustrated throughout the figures, it is envisioned that the receptacle **414** (FIG. 5) and the base **504** of the mounting structure **500** may include corresponding radiused corner portions.

(77) The front collar **506** (FIG. 5) of the mounting structure **500** is generally annular in configuration and extends outwardly from the base **504** in a forward direction. The front collar **506** includes a pair of ears **514** that extend radially outward therefrom in an integral fashion. The ears **514** are configured for releasable engagement with the cover **1000** (FIG. 4) such that the cover **1000** is connectable to and disconnectable from the image capture device **100** via the mounting structure **500**, as described in further detail below. While the front collar **506** is illustrated as including a pair of diametrically opposed ears **514** in the illustrated embodiment, it should be appreciated that the particular number, location, and/or configuration of ears **514** may be varied in alternate embodiments without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the cover **1000**).

(78) The rear collar **508** (FIGS. 6, 11) is generally annular in configuration and extends outwardly from the base **504** in a rearward direction. The rear collar **508** defines an inner transverse cross-sectional dimension (e.g., diameter) **D** and is spaced radially (laterally) inward from the apertures **512** extending through the base **504**. The rear collar **508** is configured for reception by the opening **416** (FIG. 5) to further recess the mounting structure **500** within the front housing portion **402**, thereby reducing the overall spatial requirements of the mounting structure **500**.

(79) The rear collar **508** includes an integral locating feature **516** (e.g., a detent **518** or other such projection) that is configured for receipt within a corresponding notch **802** (FIGS. 5, 6) defined by the sealing member **800** to facilitate proper relative orientation of the mounting structure **500** and the sealing member **800**. Although shown as including a single locating feature **516** in the illustrated embodiment, it should be appreciated that the particular number and/or location of the locating feature(s) **516** may varied without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the sealing member **800**).

(80) The rear collar **508** further includes a series of eyelets **520** (FIG. 11) extending radially outward therefrom that are configured for reception within corresponding reliefs **440** (FIGS. 5, 7) (e.g., cutouts, notches, indentations, etc.) defined by the flange **420** to further facilitate proper orientation of the mounting structure **500** relative to the front housing portion **402**. Each eyelet **520** defines an aperture **522** that extends partially through the mounting structure **500**. The eyelets **520** and the apertures **522** are positioned in alignment (registration) with corresponding apertures **702** (FIGS. 5, 6) in the ISLA **700** such that the ISLA **700** is directly connectable to the mounting

structure **500** via a series of (second) mechanical fasteners **442** (e.g., screws, pins, rivets, etc.) in either a fixed or removable fashion. Although shown as including three eyelets **520** and apertures **522** that are spaced (approximately) equidistant from each other (e.g., such that the apertures **522** are separated by approximately 120°), it should be appreciated that the particular number and/or location of the eyelets **520** and the apertures **522** may varied without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the ISLA **700**).

(81) The sealing member **600** (FIGS. 5, 6, 11) is located between the mounting structure **500** and the front housing portion **402**. More specifically, the sealing member **600** is positioned about the rear collar **508** of the mounting structure **500** within a corresponding channel **444** (FIG. 5) defined by the flange **420** such that the sealing member **600** extends between the apertures **512** in the base **504** and the eyelets **520**, which are received by the reliefs **440** such that the eyelets **520** extend into the channel **444**. More specifically, the sealing member **600** is positioned laterally inward of the apertures **512** and laterally outward of the eyelets **520**. To facilitate such location, in certain embodiments, such as that shown throughout the figures, the sealing member **600** may include a (first) non-annular configuration that may be characterized as generally escutcheon or shield shaped. More specifically, in the illustrated embodiment, the sealing member **600** includes a series of first segments **602ia**, **602ib**, **602ic** that are separated by a series of second segments **602iia**, **602iib**, **602iic**, whereby the segments **602ia**, **602ib** are separated by the segment **602iib**, the segments **602ib**, **602ic** are separated by the segment **602iic**, and the segments **602ia**, **602ic** are separated by the segment **602iia**. The segments **602ia**, **602ib**, **602ic** extend (generally) between the apertures **512** and each define a (generally equivalent) first curvature that generally corresponds to that of the rear collar **508** of the mounting structure **500**. The segments **602iia**, **602iib**, **602iic** are positioned adjacent to, and are configured to accommodate, the eyelets **520**. More specifically, the segments **602iia**, **602iib**, **602iic** each define a (generally equivalent) second curvature greater than the first curvature that generally corresponds to that of the eyelets **520** such that the segments **602iia**, **602iib**, **602iic** extend about the eyelets **520**. Alternative configurations for the sealing member **600**, however, would not be beyond the scope of the present disclosure. For example, an embodiment of the sealing member **600** including a generally annular configuration (e.g., an embodiment devoid of the defined segments **602ia**, **602ib**, **602ic**, **602iia**, **602iib**, **602iic** discussed above) is also contemplated herein. Additionally, while the sealing member **600** is illustrated as including a generally circular (annular) cross-sectional configuration, it should be appreciated that the particular cross-sectional configuration of the sealing member **600** may be varied in alternate embodiments without departing from the scope of the present disclosure. For example, it is envisioned that the sealing member **600** may instead include a (generally) elliptical or polygonal cross-sectional configuration.

(82) The sealing member **600** defines a (central) opening **604** for receipt of the rear collar **508** of the mounting structure **500** and the ISLA **700**. The sealing member **600** is configured to form a watertight seal between the mounting structure **500** and the front housing portion **402** and may include (e.g., may be formed partially or entirely from) any material or combination of materials suitable for this intended purpose. For example, in the illustrated embodiment, the sealing member **600** includes (e.g., is formed partially or entirely from) silicone. It should be appreciated, however, that the use of alternative material(s) of construction would not be beyond the scope of the present disclosure.

(83) Referring to FIG. 12 as well, which provides a rear, perspective view of the front housing portion illustrating connection of the ISLA **700** via the mounting structure **500**, the ISLA **700** will be discussed. The ISLA **700** (FIGS. 5, 6, 12) includes a body **704** supporting an image sensor **706** and a lens **708**, whereby the image sensor **706** and the lens **708** are integrated into a single assembly. The ISLA **700** receives and focuses light and converts captured content into an electronic image signal that is processed to form an image. The ISLA **700** extends through an opening **906** in the heatsink **900**, which allows for direct connection (securement) of the ISLA **700** to the mounting

structure **500** such that the ISLA **700** (e.g., the lens **708**) extends through the opening **416** defined by the flange **420** of the front housing portion **402**. To facilitate such connection, the ISLA **700** includes a collar **710** supporting a series of wings **712** that extend radially outward therefrom, which include the aforescribed apertures **702**. The collar **710** defines an outer transverse cross-sectional dimension (e.g., diameter) D_e (FIG. 5) and is concentrically positioned relative to the lens **708** so as to define a (generally annular) shoulder **714** that supports the sealing member **800**. Upon assembly of the image capture device **100**, the mechanical fasteners **442** are inserted through the apertures **702** in the wings **712** (in a forward direction) and into the apertures **522** in the eyelets **520** on the rear collar **508** of the mounting structure **500**, thereby directly securing together the ISLA **700** and the mounting structure **500**.

(84) Although shown as including three wings **712** that are spaced (approximately) equidistant from each other (e.g., such that the wings **712** are separated by approximately 120°), it should be appreciated that the particular number and/or location of the wings **712** may varied without departing from the scope of the present disclosure (e.g., depending upon spatial requirements, the particular configurations of the ISLA **700** and the heatsink **900**, etc.).

(85) In certain embodiments, such as that illustrated throughout the figures, it is envisioned that the heatsink **900** may include one or more recesses **908** (FIG. 5) that are configured in correspondence with the sealing member **800** (as described in further detail below) and series of reliefs **910** (e.g., cutouts, notches, indentations, etc.) that are configured in correspondence with the wings **712** of the ISLA **700**. As such, in the illustrated embodiment, the heatsink **900** includes three reliefs **910** that are spaced (approximately) equidistant from each other (e.g., such that the reliefs **910** are separated by approximately 120°). It should be appreciated, however, that the particular number and/or location of the reliefs **910** may be varied without departing from the scope of the present disclosure (e.g., depending upon the particular number and/or location of the wings **712** included on the ISLA **700**).

(86) Upon assembly of the image capture device **100**, the lens **708** of the ISLA **700** extends through the opening **909** and the wings **712** extend through the reliefs **910**, which not only facilitates proper alignment (registration) of the ISLA **700** and the heatsink **900**, but reduces the overall spatial requirements of the heatsink **900** and the ISLA **700**.

(87) By directly connecting the ISLA **700** to the mounting structure **500**, physical, supportive connections between the ISLA **700** and the heatsink **900** can be reduced, thereby simplifying assembly of the image capture device **100**. More specifically, the architecture described herein allows for a complete elimination of any physical connection between the ISLA **700** and the heatsink **900**. Additionally, connection of the ISLA **700** directly to the mounting structure **500**, rather than via an intermediate component (such as the heatsink **900**, for example), enables precise alignment of the ISLA **700** and the mounting structure **500**, which, thus, enables precise alignment of the cover **1000**, the mounting structure **500**, and the ISLA **700**. By increasing precision in the alignment of the cover **1000** and the ISLA **700** (via the mounting structure **500** located therebetween), basic functionality of the image capture device **100** can be improved. Moreover, direct connection of the ISLA **700** to the mounting structure **500** increases precision in the relative alignment of the ISLA **700** and the front housing portion **402**, which not only simplifies assembly of the image capture device **100** by reducing the number of alignments that must occur, but enhances the overall fit and finish of the image capture device **100**.

(88) The sealing member **800** (FIGS. 5, 6) is located between the mounting structure **500** and the ISLA **700**. More specifically, the sealing member **800** is positioned about the lens **708** of the ISLA **700** such that the sealing member **800** is supported by the shoulder **714** defined by the collar **710**. The sealing member **800** forms watertight seals with both the mounting structure **500** and the ISLA **700** upon assembly of the image capture device **100**, thereby obviating any need to form a direct seal between the ISLA **700** and the front housing portion **402**, and may include any material(s) suitable for this intended purpose. For example, in the particular embodiment illustrated throughout

the figures, the sealing member **800** includes (e.g., is formed partially or entirely from) conductive silicone, which facilitates grounding of the heatsink **900** relative to the mounting structure **500** (via the sealing member **800**). It should be appreciated, however, that the use of alternative material(s) of construction would not be beyond the scope of the present disclosure.

(89) The sealing member **800** includes a (second, generally annular) configuration that differs from that of the sealing member **600**. More specifically, the sealing member **800** includes a rib **804** and a plurality of feet **806** that extend radially outward from the rib **804**. The rib **804** is (generally) annular in configuration and defines a (central) opening **808** for receipt of the ISLA **700**, an inner transverse cross-sectional dimension (e.g., diameter) $D_{sub.ri}$ (FIG. 5), and an outer transverse cross-sectional dimension (e.g., diameter) $D_{sub.ro}$. The inner transverse cross-sectional dimension (e.g., diameter) $D_{sub.ri}$ is greater than the outer transverse cross-sectional dimension (e.g., diameter) $D_{sub.c}$ defined by the collar **710** of the ISLA **700** and the outer transverse cross-sectional dimension (e.g., diameter) $D_{sub.ro}$ is less than the inner transverse cross-sectional dimension D (FIG. 11) defined by the rear collar **508** of the mounting structure **500**, which allows the rib **804** to be positioned concentrically about the collar **710** and concentrically within the rear collar **508** upon assembly of the image capture device **100**. The relative dimensioning between the collar **710**, the rib **804**, and the rear collar **508** allows the sealing member **800** to simultaneously contact (engage) the ISLA **700** and the mounting structure **500**, thereby forming watertight seals with each and, thus, sealing together the ISLA **700**, the mounting structure **500**, and the front housing portion **402**. More specifically, the sealing member **800** seals with the ISLA **700** at the interface of the rib **804** and the collar **710** and with the mounting structure **500** at the interfaces between the rear collar **508**, the rib **804**, and outer (forward) surfaces **810** (FIG. 5) of the feet **806**.

(90) The feet **806** are configured for reception by and seating within the recesses **908** defined by the heatsink **900** and are circumferentially spaced from each other so as to define windows **812** therebetween. The windows **812** are positioned in alignment (registration) with the wings **712** extending from the collar **710** of the ISLA **700** such that the wings **712** and, thus, the mechanical fasteners **442**, are received by (and extend through) the windows **812** upon assembly of the image capture device **100**. While the illustrated embodiment of the sealing member **800** is shown as including three feet **806i**, **806ii**, **806iii** that are spaced (approximately) equidistant from each other (e.g., such that centerpoints thereof are separated by approximately 120°) and three windows **812i**, **812ii**, **812iii** that are spaced (approximately) equidistant from each other (e.g., such that centerpoints thereof are separated by approximately 120°), it should be appreciated that the specific number, location, and/or configuration of the feet **806** and/or the windows **812** may be varied without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the ISLA **700**).

(91) In certain embodiments, such as that seen throughout the figures, it is envisioned that the sealing member **800** may include one or more contacts **814** (FIG. 6) to facilitate grounding of the heatsink **900** relative to the mounting structure **500**. Although shown as being positioned on rear surfaces **816** of the feet **806**, it should be appreciated that the particular location of the contact(s) **814** may be varied in alternate embodiments without departing from the scope of the present disclosure.

(92) The (uppermost) foot **806ii** includes the aforementioned notch **802**, which is configured to receive the locating feature **516** (FIG. 11) extending from the rear collar **508** of the mounting structure **500** to facilitate proper relative orientation of the mounting structure **500** and the sealing member **800**. Although shown as including a single notch **802** in the illustrated embodiment, it should be appreciated that the particular number and/or location of the notch(es) **802** may be varied without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the mounting structure **500**). For example, it is envisioned that each foot **806** may define a notch **802** that is configured to receive a corresponding locating feature **516** extending from the rear collar **508**.

(93) With reference now to FIG. 4 and FIG. 13, the cover **1000** will be discussed. FIG. 13 provides a front, perspective view of the housing assembly with the cover **1000** connected to the mounting structure **500**. The cover **1000** functions as a removable cap that protects and conceals the ISLA **700** and includes a frame **1002** and an insert **1004** that is supported by the frame **1002**. The insert **1004** includes (e.g., is formed partially or entirely from) an optically clear material so as to permit image capture through the cover **1000** without interference. Although shown as being generally square-shaped in configuration throughout the figures, it should be appreciated that the specific configuration of the cover **1000** may be varied in alternate embodiments without departing from the scope of the present disclosure (e.g., depending upon the particular configuration of the ISLA **700** and/or the front housing portion **402**, the desired aesthetic appearance of the image capture device **100**, etc.).

(94) The cover **1000** and the mounting structure **500** are configured for direct connection. More specifically, the cover **1000** is configured for removable connection to the ears **514** extending radially outward from the front collar **506** of the mounting structure **500**. For example, the cover **1000** may include one or more pockets (or other such recesses or openings) that are configured in correspondence with the ears **514**. To facilitate connection and disconnection of the cover **1000**, it is envisioned that the ears **514** may include (or otherwise define) angled (e.g., chamfered) bearing surfaces **524** that are configured to urge the cover **1000** outwardly (away from the front housing portion **402**) upon rotation of the cover **1000**.

(95) Directly connecting the cover **1000** to the mounting structure **500** allows for the elimination of any interposed components (e.g., the heatsink **900** (FIGS. 5, 6)) between the cover **1000**, the mounting structure **500**, and the front housing portion **402**. The elimination of interposed components between the cover **1000**, the mounting structure **500**, and the front housing portion **402** reduces the number of components that must be properly aligned during assembly of the image capture device **100**, which not only simplifies the assembly process, but improves the overall alignment of the cover **1000** relative to the front housing portion **402** and the ISLA **700** (FIGS. 5, 6), thereby enhancing the overall fit and finish of the image capture device **100**.

(96) To facilitate sealing between the cover **1000** and the mounting structure **500**, the image capture device **100** includes a sealing member **1100** (e.g., an O-ring **1102**) that is positioned between the cover **1000** and the mounting structure **500**. More specifically, upon connection of the cover **1000** to the mounting structure **500**, the sealing member **1100** is positioned within a corresponding (generally annular) groove **1006** defined by the cover **1000** and is received by the front collar **506** such that the sealing member **1100** is concentrically positioned within the front collar **506** so as to form a watertight seal between the cover **1000** and the mounting structure **500**. While the sealing member **1100** is illustrated as including a generally circular (annular) cross-sectional configuration, it should be appreciated that the particular configuration of the sealing member **1100** may be varied in alternate embodiments without departing from the scope of the present disclosure.

(97) To further enhance sealing of the image capture device **100**, it is envisioned that the image capture device **100** may include a variety of additional sealing members. For example, as seen in FIGS. 14 and 15, which provide front, perspective views of the image capture device **100**, the image capture device **100** may include a sealing member **1200** that is configured to form a watertight seal between the front housing portion **402** and the door **114**; a sealing member **1300** that is configured to form a watertight seal about the light pipe; a sealing member **1400** that is configured to form a watertight seal about one or more (top, upper) microphones; a sealing member **1500** that is configured to form a watertight seal about one or more (side, lateral) microphones; a sealing member **1600** that is configured to form a watertight seal about one or more vents that allow for pressure equalization with the external environment by allowing air to pass therethrough; and a sealing member **1700** that is supported by an external (e.g. lower) surface of the front housing portion **402** (rather than being positioned internally) so as to form a watertight seal about one or more speakers. The image capture device **100** may further include a sealing member **1800**

that is configured to form a watertight seal about one or more (front and/or rear) microphones. As seen in FIG. 14, for example, the sealing member **1800** includes an array of openings **1802** (e.g., arranged in a 3×3 pattern), rather than a single opening **1802**, to thereby improve acoustics.

(98) To facilitate sealing between the heatsink **900** (FIGS. 5, 6) and the front housing portion **402** it is envisioned that the image capture device **100** may also include one or more sealing members **1900**. For example, as seen in FIG. 15, the sealing member(s) **1900** may be configured and positioned to form a watertight seal about the interconnect mechanism **136** (FIG. 1B), which may allow for direct connection of the interconnect mechanism **136** to the heatsink **900**. Although shown as including four sealing members **1900** that are generally annular in configuration (e.g., O-rings **1902**), it should be appreciated that the particular number, location, and/or configuration of the sealing member(s) **1900** may be varied in alternate embodiments without departing from the scope of the present disclosure. By including separate, discrete sealing members **1900**, it is envisioned that assembly of the image capture device **100** may be simplified by obviating the need for a larger seal that may otherwise be required to seal the perimeter of the heatsink **900** to the front housing portion **402**.

(99) With reference to FIGS. 16 and 17, which provide front and rear perspective views of the image capture device **100**, respectively, the image capture device **100** may further include a (front) sealing member **2000** and a (rear) sealing member **2100**. The sealing members **2000**, **2100** are configured in correspondence with the display **108** and the interactive display **138**, respectively, to thereby form a watertight seal between the display **108** and the front housing portion **402** and a watertight seal between the interactive display **138** and the rear housing portion **404**.

(100) With reference now to FIGS. 4-6, assembly of the image capture device **100** will be discussed. Initially, the mounting structure **500** is (externally) connected to the front housing portion **402** via the mechanical fasteners **438**, which are inserted rearwardly. More specifically, the base **504** is positioned adjacent to the flange **420** such that the sealing member **600** is positioned and compressed therebetween so as to form a watertight seal between the front housing portion **402** and the mounting structure **500**. Thereafter, the light pipe, the drain cover, etc., and the heatsink **900** are connected (secured) to the front housing portion **402**. The GPS board, the main board, the battery cage, etc., are then connected to the heatsink **900** and any other necessary electrical connections are made.

(101) With the heatsink **900** in place, the sealing member **800** is brought into contact (engagement) with the mounting structure **500** by positioning the sealing member **800** such that the rib **804** is received by the rear collar **508**. Thereafter, the ISLA **700** is positioned such that the lens **708** extends through the heatsink **900** (via the opening **906**), through the sealing member **800**, through the front housing portion **402**, through the sealing member **600**, and through the mounting structure **500**. When so positioned, the collar **710** of the ISLA **700** is received by the rib **804** of the sealing member **800** and the wings **712** are received by the windows **812** defined between the feet **806**. The ISLA **700** is then connected to the mounting structure **500** via forward insertion of the mechanical fasteners **442** through the apertures **702** in the wings **712**, through the sealing member **800**, and into the apertures **522** in the eyelets **520** of the rear collar **508** of the mounting structure **500**, thereby compressing the sealing member **800** between the ISLA **700** and the mounting structure **500** and sealing together the ISLA **700**, the mounting structure **500**, and the front housing portion **402** via the sealing members **600**, **800**.

(102) Following connection of the ISLA **700** to the mounting structure **500**, the interactive display **138** (FIG. 1B) is connected (secured) and sealed to the rear housing portion **404** and the adhesive **434** (FIG. 10) is applied to the recess **432** defined by the rear housing portion **404** and/or the tongue **424** extending from the front housing portion **402**. Before the adhesive **434** dries (cures), the rear housing portion **404** is electrically connected to the heatsink **900** (e.g., via the FPC(s) **426** (FIG. 10)) and the respective front and rear housing portions **402**, **404** are connected via insertion of the tongue **424** within the recess **432**. The adhesive **434** is then allowed to dry (cure) to connect

(secure) together the housing portions **402**, **404** and form a seal therebetween, establishing a watertight internal environment for the components of the image capture device **100**. The interconnect mechanism **136** (FIG. **1B**) can then be connected and the cover **1000** can be connected to the mounting structure **500** via the ears **514** on the front collar **506**.

(103) In certain embodiments, the ISLA includes at least one optical element and at least one image sensor.

(104) In certain methods, connecting the ISLA to the mounting structure includes positioning the second fasteners radially inward of the first fasteners.

(105) In certain embodiments, connecting the ISLA to the mounting structure includes inserting the ISLA through the mounting structure and through the front housing portion of the image capture device.

(106) While the disclosure has been described in connection with certain embodiments, it is to be understood that the disclosure is not to be limited to the disclosed embodiments but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

(107) Persons skilled in the art will understand that the various embodiments of the disclosure described herein, and shown in the accompanying figures, constitute non-limiting examples, and that additional components and features may be added to any of the embodiments discussed hereinabove without departing from the scope of the present disclosure. Additionally, persons skilled in the art will understand that the elements and features shown or described in connection with one embodiment may be combined with those of another embodiment without departing from the scope of the present disclosure to achieve any desired result, and will appreciate further features and advantages of the presently disclosed subject matter based on the description provided.

Variations, combinations, and/or modifications to any of the embodiments and/or features of the embodiments described herein that are within the abilities of a person having ordinary skill in the art are also within the scope of the disclosure, as are alternative embodiments that may result from combining, integrating, and/or omitting features from any of the disclosed embodiments.

(108) Use of the term “optionally” with respect to any element of a claim means that the element may be included or omitted, with both alternatives being within the scope of the claim.

Additionally, use of broader terms such as “comprises,” “includes,” and “having” should be understood to provide support for narrower terms such as “consisting of,” “consisting essentially of,” and “comprised substantially of.” Accordingly, the scope of protection is not limited by the description set out above, but is defined by the claims that follow, and includes all equivalents of the subject matter of the claims.

(109) In the preceding description, reference may be made to the spatial relationship between the various structures illustrated in the accompanying drawings, and to the spatial orientation of the structures. However, as will be recognized by those skilled in the art after a complete reading of this disclosure, the structures described herein may be positioned and oriented in any manner suitable for their intended purpose. Thus, the use of terms such as “above,” “below,” “upper,” “lower,” “inner,” “outer,” “left,” “right,” “upward,” “downward,” “inward,” “outward,” “horizontal,” “vertical,” etc., should be understood to describe a relative relationship between the structures and/or a spatial orientation of the structures. Those skilled in the art will also recognize that the use of such terms may be provided in the context of the illustrations provided by the corresponding figure(s).

(110) Additionally, terms such as “approximately,” “generally,” “substantially,” and the like should be understood to allow for variations in any numerical range or concept with which they are associated. For example, it is intended that the use of terms such as “approximately” and “generally” should be understood to encompass variations on the order of 25%, or to allow for manufacturing tolerances and/or deviations in design.

(111) Although terms such as “first,” “second,” etc., may be used herein to describe various operations, elements, components, regions, and/or sections, these operations, elements, components, regions, and/or sections should not be limited by the use of these terms in that these terms are used to distinguish one operation, element, component, region, or section from another. Thus, unless expressly stated otherwise, a first operation, element, component, region, or section could be termed a second operation, element, component, region, or section without departing from the scope of the present disclosure.

(112) Each and every claim is incorporated as further disclosure into the specification, and represents embodiments of the present disclosure. Also, the phrases “at least one of A, B, and C” and “A and/or B and/or C” should each be interpreted to include only A, only B, only C, or any combination of A, B, and C.

Claims

1. A method of assembling an image capture device, the method comprising: connecting a mounting structure to a front housing portion of the image capture device via first fasteners, wherein the mounting structure includes: a base; a front collar extending from the base in a forward direction; and a rear collar extending from the base in a rearward direction; connecting an integrated sensor-lens assembly (ISLA) to the mounting structure via second fasteners, wherein the ISLA includes at least one optical element and at least one image sensor; and connecting a rear housing portion of the image capture device to the front housing portion.
2. The method of claim 1, wherein connecting the mounting structure to the front housing portion further includes: connecting the mounting structure to an exterior surface thereof.
3. The method of claim 2, wherein connecting the ISLA to the mounting structure includes inserting the ISLA into the front housing portion in a forward direction.
4. The method of claim 3, wherein connecting the ISLA to the mounting structure further includes: positioning the ISLA such that the ISLA extends through the mounting structure.
5. The method of claim 4, wherein positioning the ISLA includes inserting the ISLA through an opening defined by the base.
6. The method of claim 1, wherein connecting the mounting structure to the front housing portion includes inserting the first fasteners into first apertures positioned radially outward of the front collar.
7. The method of claim 6, wherein connecting the ISLA to the mounting structure includes inserting the second fasteners into second apertures positioned radially inward of the first apertures.
8. The method of claim 1, further comprising: positioning a first sealing member between the mounting structure and the front housing portion to form a watertight seal therebetween; and positioning a second sealing member between the mounting structure and the ISLA to form a watertight seal therebetween.
9. The method of claim 8, wherein positioning the first sealing member and positioning the second sealing member includes supporting the first sealing member and the second sealing member by the mounting structure.
10. The method of claim 1, wherein connecting the ISLA to the mounting structure includes positioning the second fasteners radially inward of the first fasteners.
11. The method of claim 1, wherein connecting the ISLA to the mounting structure includes inserting the ISLA through the mounting structure and through the front housing portion of the image capture device.
12. A method of assembling an image capture device, the method comprising: connecting a mounting structure to an exterior surface of a front housing portion of the image capture device; connecting an integrated sensor-lens assembly (ISLA) to the mounting structure such that the ISLA is positioned internally within the front housing portion; and connecting a rear housing portion of

the image capture device to the front housing portion.

13. The method of claim 12, wherein connecting the mounting structure to the exterior surface of the front housing portion includes advancing the mounting structure in a first direction.

14. The method of claim 13, wherein connecting the mounting structure to the exterior surface of the front housing portion further includes: positioning the mounting structure such that the mounting structure extends into the front housing portion.

15. The method of claim 14, wherein connecting the ISLA to the mounting structure includes advancing the ISLA in a second direction opposite to the first direction.

16. The method of claim 15, wherein connecting the mounting structure to the exterior surface of the front housing portion further includes: inserting first fasteners into the mounting structure, and wherein connecting the ISLA to the mounting structure further includes: inserting second fasteners into the mounting structure such that the second fasteners are positioned radially inward of the second fasteners.

17. A method of assembling an image capture device, the method comprising: positioning a first sealing member between a body of the image capture device and a mounting structure; externally connecting the mounting structure to the body of the image capture device such that the first sealing member is positioned therebetween; positioning a second sealing member between the mounting structure and an integrated sensor-lens assembly (ISLA); and internally connecting the ISLA to the mounting structure by inserting the ISLA through the mounting structure and through the body of the image capture device.

18. The method of claim 17, wherein externally connecting the mounting structure to the body of the image capture device includes inserting first fasteners into the mounting structure.

19. The method of claim 18, wherein internally connecting the ISLA to the mounting structure includes inserting second fasteners into the mounting structure such that the second fasteners are positioned radially inward of the first fasteners.

20. The method of claim 19, wherein internally connecting the ISLA to the mounting structure further includes: inserting the ISLA through a heatsink of the image capture device.
